

Coding & Solution

Date	15 March 2026
Team ID	SWTID-2026-5650
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	5 Marks

Coding & Solution:

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary:

Field	Details
Repository Link / URL	https://github.com/SWTID-2026-5650/ComicCraft-AI-Comic-Creator
Programming Language(s)	Python 3.11, JavaScript (React / Streamlit Frontend)
Framework(s) Used	Streamlit, Google Generative AI SDK (Gemini 1.5 Flash/Pro), PIL (Pillow), ReportLab
Key Features Implemented	AI Story & Character Generator, Dynamic Panel Scripting, Text-to-Comic Panel Art Generation, Speech Bubble Overlay, PDF/PNG Export
Pending / Incomplete Features	None (All core and advanced features fully operational)
Setup / Run Instructions	1. Clone repository 2. pip install -r requirements.txt 3. Set GEMINI_API_KEY in .env 4. Run streamlit run app.py

Code Quality Checklist:

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes

S.No	Criteria	Status (Yes / No)
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

ComicCraft leverages Google Gemini 1.5 Models to generate multi-panel comic narratives, character visual descriptors, and dialogue scripts. It automatically coordinates text generation with image prompts and layout engines to produce complete, exportable comic books. The solution achieves robust performance through async API orchestration, automated asset composition, and graceful error handling.